ROHIT MAITY

BACKEND / AI ENGINEER

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Third-year CSE student (CGPA 9.02) who builds **production-grade agentic AI systems and backends** — LangGraph agents with checkpointed state, durable job queues, circuit-broken LLM providers and realtime streaming services. Python-first, with depth in **agent design, distributed systems and API security**. Smart India Hackathon 2025 National Finalist.

PROJECTS

Walkthru

GitHub

AI launch-readiness testing agent · FastAPI · LangGraph · PostgreSQL · MCP

- Built a **LangGraph persona agent** that runs real user journeys in the user's own Chrome through a **stateless FastAPI API** — one HTTP call per step via `interrupt()` / `Command(resume)` and a Postgres checkpointer, so the API scales horizontally.
- Enforced safety **in code, not prompts**: a server-side gate re-checked every step replaces hallucinated elements and safe-stops pay/delete actions; an SSRF-safe fetcher blocks private ranges and re-validates every redirect.
- Engineered for failure: durable **Postgres job queue** (`SKIP LOCKED`, leases, jittered backoff), circuit-broken multi-provider **LLM failover**, and a 24-tool remote **MCP server** with hashed, rate-limited keys.
- Verified by **1,185 API tests** and LangSmith evals on adversarial sites; the readiness score is computed in code, never by the model.

TripVerse

GitHub · Live

Agentic travel planner · LangGraph · FastAPI · PostgreSQL · SQLAlchemy / asyncpg · React

- Designed two **LangGraph planning graphs** (researched itinerary and day-by-day copilot) streamed over FastAPI into one normalized trip document that drives the plan, map, 3D route, budget ledger and export.
- **"LLMs propose, code decides"** — budget, routing cost, day capacity and ranking are deterministic, tested functions; every edit returns a change receipt and replies are checked against computed facts.
- Rate-limit-aware multi-provider failover with deterministic fallbacks for extraction and search; **188 backend tests**.

SkyGuide AI

GitHub · Live

Distributed realtime astronomy system · FastAPI · Express · Socket.IO · MongoDB

- Architected **3 independent services**; a private FastAPI science engine is reachable only through an allowlisted reverse proxy that attaches a shared key server-side.
- Scores **13,311 deep-sky objects** per request in one vectorised Astropy / NumPy transform, cached with stale-while-revalidate in MongoDB (TTL index, single-flight de-duplication).
- Live telescope guidance over a **20 Hz WebSocket stream** — ephemeris extrapolation (<0.01° error) and a hysteresis lock state machine; layered rate limiting and JWT auth.

KOE

GitHub · Live

Voice-first workday agent · Python · FastAPI · PostgreSQL · WebSockets · OAuth · React

- Built live voice calls, chat and voice notes over a single **FastAPI tool registry** integrating GitHub, Calendar, Docs, Sheets, Notion and Spotify.
- Persisted **human-in-the-loop approval gate**: exact-argument proposals in Postgres with 10-minute expiry and an atomic pending-to-executing claim, so a double confirmation can never run a write twice.
- **Realtime audio pipeline** — 16 kHz PCM16 capture, gapless 24 kHz playback, interruption-driven queue flush and session resumption; encrypted OAuth tokens.

SKILLS

LANGUAGES

Python (advanced) · JavaScript · Java · C · SQL

AGENTIC AI & SYSTEMS

LangGraph · LangSmith evals · RAG · tool calling · MCP · prompt & context engineering · agent memory · HITL

ML & DEEP LEARNING

PyTorch · Transformers & attention · sequence models (RNN/LSTM) · generative models (VAE, GAN, DDPM) · scikit-learn · NumPy · Pandas

BACKEND

FastAPI · Pydantic · async Python · Node.js · Express · REST · WebSockets · JWT / OAuth

SYSTEMS & ARCHITECTURE

Distributed systems · realtime streaming · job queues · caching · rate limiting · circuit breakers · system design

DATABASES

PostgreSQL · MongoDB · Redis · MySQL · Supabase (RLS)

FRONTEND & TOOLS

React · HTML / CSS · Git · Linux

ACHIEVEMENTS

SIH 2025 — National Finalist
Smart India Hackathon (PS 25007). Built the *LivestockIQ* backend — Node / Express / MongoDB, JWT, PDFKit reporting for AMU/MRL monitoring.

SBI Ideathon 2025 — Top 1000
Technical Lead on *PraanaCare* (AI telemedicine) — owned architecture, stack selection and system design.

EDUCATION

Heritage Institute of Technology, Kolkata

B.Tech, Computer Science & Engineering (IoT, Cybersecurity & Blockchain) · 2024–2028

`CGPA 9.02 / 10 · no backlogs`

DSA · OOP · DBMS · Operating Systems · Computer Networks

`Class XII 90% · Class X 96% (State Rank 20)`